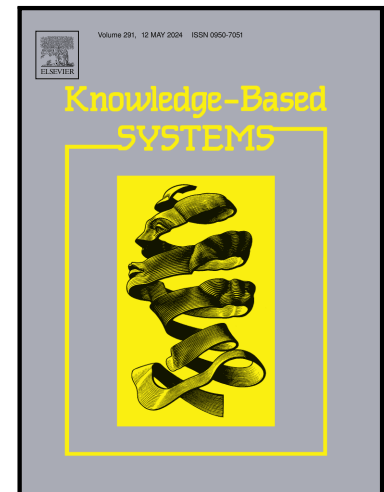


SFD-ViT: Spatial-Frequency Dual-Stream Transformer with Patch Embedding for Face Forgery Detection

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1 Highlights

2 **SFD-ViT: Spatial-Frequency Dual-Stream Transformer with Patch** 3 **Embedding for Face Forgery Detection**

4 Yikang Yang, Yulin Deng, Qiuxu Yi, Jun Yi*

- 5 • A spatial-frequency dual-stream Vision Transformer for face forgery
6 detection
- 7 • Adaptive fusion of spatial and frequency features to focus on crucial
8 forgery traces
- 9 • Superior performance across multiple face forgery datasets over existing
10 methods
- 11 • Comprehensive ablation studies validate contribution of each compo-
12 nent

SFD-ViT: Spatial-Frequency Dual-Stream Transformer with Patch Embedding for Face Forgery Detection

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Abstract

The rapid advancement of generative models, such as generative adversarial networks and diffusion models, has enabled the synthesis of highly realistic forged faces, posing significant threats to security and privacy. Consequently, developing robust face forgery detection algorithms has become imperative. In this work, we propose a novel Spatial-Frequency Dual-Stream Vision Transformer (SFD-ViT) that **revealing subtle forgery artifacts from both spatial and frequency domains**. Specifically, our framework incorporates a pre-trained EfficientNetV2-S backbone for feature extraction, a Squeeze-and-Excitation module to enhance discriminative representations, and a Transformer module with Patch Embedding to capture global inconsistencies. Extensive experiments on multiple face forgery datasets demonstrate that the SFD-ViT consistently outperforms baseline methods, exhibiting strong performance across diverse manipulation techniques. Overall, this work provides an effective solution to safeguard face detection systems against emerging forgery threats.

Keywords: Face Forgery Detection, Multi-Modal Fusion, EfficientNet, Vision Transformer

1. Introduction

1.1. Problem definition

Image serves as an crucial medium of information transmission, its authenticity and security has become the central concerns in society [1]. The maliciously forged images pose a risk of unauthorized dissemination, which threatens personal privacy, commercial interests and public security [2]. As a distinctive and pivotal form of images, the human face serving as biometric characteristic and has been widely used in identity authentication or payment systems [3, 4]. Nonetheless, as an important component of personal privacy data, unauthorised use of facial images may lead to identity theft, fraudulent behavior, and even threats the security of personal property [5]. Therefore, with the increasing prevalence of facial detection technology, the assurance of its security and reliability has become a prevalent subject of concern within both academy and industry.

Generative models produces highly similar new data by learning data distributions [6]. They have demonstrated powerful capabilities in image synthesis, text generation and audio processing [7, 8]. Hence, generative models have made it possible to generate realistic forged face images that are difficult to distinguish from real ones [9, 10, 11, 12]. While some forged faces are created for entertainment, such actions still potentially impacting individual reputation and privacy. More seriously, maliciously forged face may be used to commit illegal or criminal activities such as financial fraud and identity theft [13, 14]. Therefore, developing efficient and robust face forgery detectors is an important guarantee for maintaining social security.

1.2. Existing problems

The rapid evolution of face forgery generation techniques has prompted considerable research endeavours to develop robust detectors, but innovation

is still needed to deal with constantly evolving generative models [15]. Existing approaches primarily focus on leveraging visible artifacts introduced during the forged face creation, such as inconsistencies in facial geometry or texture irregularities. This makes it difficult to comprehensively capture the multi-domain forgery traces, resulting in the poorer performance while detecting high quality generated images. Consequently, fully exploring the complementarity of multi-domain information is necessary for the model to learn more forged artifacts. However, multi-domain information (such as spatial and frequency domain) exhibits different characteristics, making the integration of important features remains a crucial problem to be solved. Based on these background and issues, we conduct this research, hoping to resist the influence of increasingly advanced face forgery methods.

1.3. Major contributions

The major contributions of this work are summarized as follows:

- We propose a novel Spatial-Frequency Dual-Stream Vision Transformer (SFD-ViT) architecture. Our model processes both RGB images and their frequency representations (via Discrete Cosine Transform, DCT) in parallel, enabling it to capture a more comprehensive forgery traces.
- We design an adaptive feature enhancement pipeline. We extract features by the lightweight backbone EfficientNetV2-S (ENV2S), then employ a Squeeze-and-Excitation (SE) module in each stream. The SE modules highlight critical patterns and suppress trivial features, thus focusing on features that are more relevant to classification.
- We introduce a fusion block based on the Transformer with Patch Embedding to capture long-range dependencies of the dual-stream features. This allows the model to identify global inconsistencies of face forgery, which are often missed by local-receptive-field models.

- We conduct extensive experiments on multiple datasets encompassing various forgery techniques. The results demonstrate that our SFD-ViT outperforms existing baseline models. Ablation study further rigorously validates the effectiveness of each component.

The arrangement of the paper is shown as follows. Section 2 introduces the related works. Section 3 is the detail of the proposed model. In Section 4, we describe datasets, experimental setup and results. In Section 5, we present the ablation study and results. The summary and Discussion are presented in Section 6. Finally, we conclude this work in Section 7.

2. Related works

The deep generative models can be maliciously used to create realistic forged face images and videos, which erode the public trust [16]. Consequently, the development of robust and generalizable face forgery detectors has become an urgent and critical challenge [17]. In this section, we provide a brief introduction to existing related works.

2.1. Spatial-domain detection

The edges, textures, colors, and other visible characteristics are intuitive features for images, making it possible to track face forgery traces in spatial domain [18]. Liao et al. [19] locate the facial landmarks in the video and constructs facial muscle movement, then integrate forensic knowledge to complete the detection. At present, the hybrid architectures is a subject of interest to researchers [20]. Al-Dulaimi et al. [21] introduce the hybrid convolutional neural network (CNN)-long short-term memory (LSTM) approach that merging spatial awareness and temporal context understanding. Yang et al. [22] capture forgery traces by adopting pre-training on image transformations and patchwise contrastive learning. The dual-stream structure also shows its potential in face forgery detection. Shu et al. [23] present a

R4Q1

103 dual-stream CNN that **employing** multi-scale color inversion and RGB inputs
 104 to extract reflection and color features. Yadav et al. [24] introduce an **adap-**
 105 **tive** weighting mechanism that dynamically combines manipulation residuals
 106 and texture features. Guo et al. [25] design tensor preprocessing and ma-
 107 nipulation trace attention modules, then combine the gradient operator to
 108 recognize face forgery.

109 2.2. Frequency-domain detection

110 Considering spatial-domain features cannot fully catch forgery traces,
 111 the analysis of frequency-domain **traces** has emerged as a promising direction
 112 [26]. Zhou et al. [27] focus on **middle-high** frequency **traces** through fine-
 113 grained enhancement modules for robust detection. Convertini et al. [28]
 114 propose a Discrete Fourier Transform (DFT)-based face forgery detector,
 115 **the** frequency features are classified by the random forest. Tan et al. [29]
 116 propose a frequency-aware approach, which focuses on the high frequency
 117 representation of features across spatial and channel dimensions.

118 2.3. Multiple domain detection

119 Generative models leave **traces** in both spatial and frequency domains.
 120 Hence, multi-domain information may improve the recognition performance
 121 [30]. Zhao et al. [31] design a **dual-stream** framework that concatenate
 122 the Transformer-based noise tracking and CNN-based multi-level frequency-
 123 aware modules. Kiruthika et al. [32] leverage images quality assessment
 124 features from spatial and frequency domains to recognize the forged face.
 125 Liang et al. [33] propose a **dual-stream** network with hierarchical supervi-
 126 sions, employ **simple fusion methods** to utilize spatial and frequency streams
 127 to handle low-quality forged faces. However, the effective fusion of multiple
 128 domain features remains a key challenge. Qiu et al. [34] integrate information
 129 from both the spatial and frequency domain through feature superposition
 130 strategy. Sun et al. [35] develop a cross-domain fusion method incorporating

frequency-domain edge geometric features. Wang et al. [36] extract shallow features from spatial stream, obtain amplitude and phase features from frequency stream, after that fuse them for classification. Liang et al. [37] propose a CNN-LSTM model **integrating** with a facial geometry prior module to exploit geometric and frequency-domain features for pixel-level manipulation localization. These approaches collectively demonstrate the critical role of multi-domain analysis in exposing subtle forgery patterns.

2.4. Comparisons between our model and others

The **face forgery detection** still represents a challenge for researchers. Firstly, face forgery techniques are undergoing rapid advancement, various methods generating images of an increasingly nature. It decreases the efficacy of existing single-domain models in fully capturing forgery **traces**. Secondly, although multi-domain information is readily available, the effective integration of these information to emphasize discriminative features while suppressing noise **still** remains an open challenge. To address these issues, the SFD-ViT is proposed for **face forgery** detection, **and the analysis of differences between our approach and several related works is presented below**.

MRT-Net [24] uses residue and textual stream to extract features from spatial domain, while we obtain information from spatial and frequency domain. ANTEM [31] consists of an adaptive noise trace enhancement module and a multi-level frequency aware module, **but we use the SE module to focus on important features**. IQA-based model [32] also employs a multi-stream architecture, but it concatenates the features, then utilize the random forest as the classifier, which is completely different from our model. HTNet [33] utilizes **the** RGB and DCT-transformed images, but it employs summarization, concatenation, and channel shuffle as fusion schemes, while we use Transformer **with Patch Embedding**. MC-LCR [36] employs the frequency-domain information, but it extracts features from grey image patches and uses DFT

159 to obtain the frequency information, while we employ the DCT. In summary,
 160 our method is different from existing approach, and we will provide a detailed
 161 description of the proposed method in the following section.

162 3. Methodology

163 The proposed SFD-ViT will be described in this Section. Firstly, the
 164 overall architecture will be introduced, followed by a detailed presentation of
 165 the model in Section 3.2 to 3.5.

166 3.1. Architecture Overview

167 In response to the growing demand for face forgery detectors, we pro-
 168 pose a model named SFD-ViT. Firstly, we introduce both RGB images and
 169 DCT-transformed frequency-domain images, the latter of which help to iden-
 170 tify potential face forgery traces in the frequency domain and compensate
 171 for the drawbacks of using only spatial domain. Subsequently, these images
 172 will be input into feature extraction modules containing ENV2S and SE. The
 173 former employs the pretrained model that removed classifier to extract fea-
 174 tures, while the latter adaptively recalibrates features and enables the model
 175 to focus on key information. After that, the extracted features will be input
 176 into fusion block including Patch Embedding and Transformer. The former
 177 performs image segmentation, flattening, and positional encoding to obtain
 178 high-dimensional features, while the latter understands the relationships be-
 179 tween image blocks and extracts global information for classification. Finally,
 180 this information is fed into a fully-connected layer to classify face images.

181 The architecture of proposed network is shown as Fig. 1.

182 3.2. Image processing

183 In this part, we utilize both the raw RGB image and its frequency con-
 184 version to extract features from images.

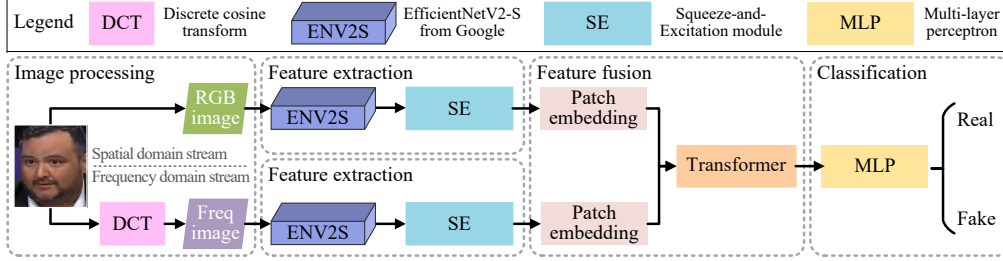


Fig. 1: The proposed network architecture.

185 3.2.1. *Spatial-domain stream*

186 In the **spatial-domain** stream, we employ the RGB images to capture
 187 original color and texture features by inputting them into the ENV2S.

188 3.2.2. *Frequency-domain stream*

189 In the **frequency-domain** stream, we utilize the DCT to decompose im-
 190 ages into the frequency domain and obtain the frequency details [38].

191 For an image $f(a, b)$ in size $m \times n$, the DCT can be obtained as:

$$D(u, v) = c(u)c(v) \sum_{a=0}^{m-1} \sum_{b=0}^{n-1} f(a, b) \times \cos \left[\frac{(a + 0.5) \pi}{m} u \right] \cos \left[\frac{(b + 0.5) \pi}{n} v \right], \quad (1)$$

192 where, $D(u, v)$ is the DCT coefficient of image $f(a, b)$, $u = 0, 1, \dots, m - 1$,
 193 $v = 0, 1, \dots, n - 1$, $c(u)$ and $c(v)$ can be obtained as:

$$c(u) = \begin{cases} \sqrt{\frac{1}{m}}, u=0 \\ \sqrt{\frac{2}{m}}, u \neq 0 \end{cases}, c(v) = \begin{cases} \sqrt{\frac{1}{n}}, v=0 \\ \sqrt{\frac{2}{n}}, v \neq 0 \end{cases}. \quad (2)$$

195 3.3. *Feature extraction*

196 In this part, we employ the ENV2S and SE modules to extract features
 197 from spatial and frequency domains. The ENV2S is a **lightweight network**

that effectively extracts the high-level semantic features of images. The SE module adaptively recalibrates features, thereby highlighting important features and suppressing redundant information. The architecture of feature extraction is shown in Fig. 2.

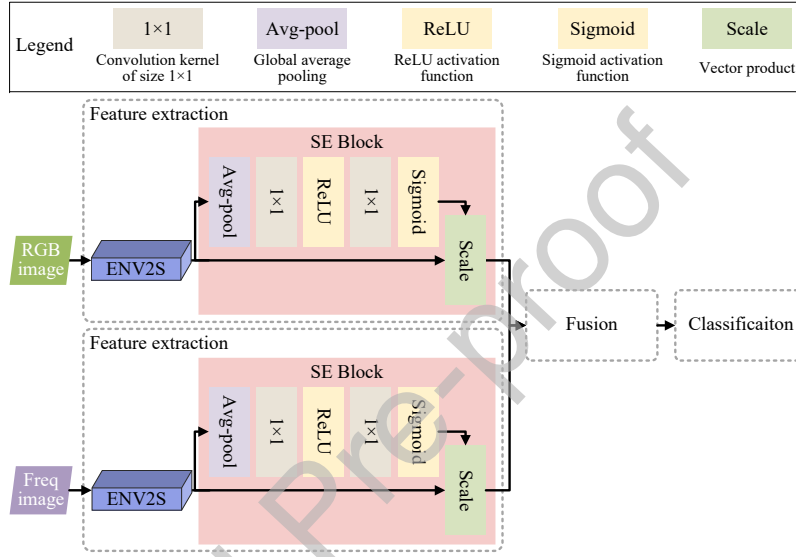


Fig. 2: The architecture of feature extraction.

3.3.1. Spatial-domain stream

1) ENV2S backbone

The original classifier of the ENV2S is deleted and its pretraining weight is utilised to extract the high-level feature representation of RGB images.

In the **spatial-domain** stream, we extract the feature by ENV2S:

$$\mathbf{F}_{rgb} = \text{EN}(\mathbf{X}), \mathbf{F}_{rgb} \in \mathbb{R}^{B \times C \times 7 \times 7}, \quad (3)$$

where $\mathbf{X} \in \mathbb{R}^{B \times 3 \times 224 \times 224}$ is the input image, B is the batch size, $C = 1280$ is the number of channels, EN is the ENV2S and the details are shown in [39].

2) SE module

210 To further enhance the channel attention, the SE module is employed
 211 on the processing of the features obtained by the ENV2S.

212 Firstly, **we obtain** the global channel statistics by adaptive average pool-
 213 ing:

$$\mathbf{z}_{rgb} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_{rgb(c,i,j)}, \mathbf{z}_{rgb} \in \mathbb{R}^{B \times C}, \quad (4)$$

214

215 where $\mathbf{F}_{rgb(c,i,j)}$ is the feature map of the c -th channel, H and W are the
 216 height and width of the feature map, respectively.

217 Subsequently, **we generate** the attention by convolution and activation
 218 functions:

$$\mathbf{s}_{rgb} = \sigma_S(\mathbf{W}_2 \cdot \sigma_R(\mathbf{W}_1 \cdot \mathbf{z}_{rgb})), \mathbf{s}_{rgb} \in \mathbb{R}^{B \times C}, \quad (5)$$

219 where $\mathbf{W}_1$ and $\mathbf{W}_2$ is the first and second fully-connected weight, $\mathbf{W}_1 \in$
 220 $\mathbb{R}^{C \times C/r}$, $\mathbf{W}_2 \in \mathbb{R}^{C/r \times C}$, $r = 16$ is the compression rate, σ_S and σ_R is the
 221 Sigmoid and ReLU activation function, respectively.

222 After that, **we reweight** features to highlight the salient features and
 223 suppress the noise:

$$\tilde{\mathbf{F}}_{rgb} = \mathbf{s}_{rgb} \cdot \mathbf{F}_{rgb}, \tilde{\mathbf{F}}_{rgb} \in \mathbb{R}^{B \times C \times 7 \times 7}. \quad (6)$$

224 3.3.2. Frequency-domain stream

225 1) ENV2S

226 The process of frequency-domain images is similar to that of RGB im-
 227 ages:

$$\mathbf{F}_{freq} = \text{EN}(\mathbf{D}), \mathbf{F}_{freq} \in \mathbb{R}^{B \times C \times 7 \times 7}. \quad (7)$$

228 2) SE module

229 The process of SE module in frequency-domain stream is similar to that
 230 **of** spatial domain, and the features can be obtained as:

$$\tilde{\mathbf{F}}_{freq} = \mathbf{s}_{freq} \cdot \mathbf{F}_{freq}, \tilde{\mathbf{F}}_{freq} \in \mathbb{R}^{B \times C \times 7 \times 7}. \quad (8)$$

231 3.4. Feature fusion

232 In this part, the **Transformer with Patch Embedding** is employed to
 233 fuse features of the spatial and frequency domain, which **captures** the global
 234 context and long-range dependencies. The architecture of feature fusion is
 shown in Fig. 3.

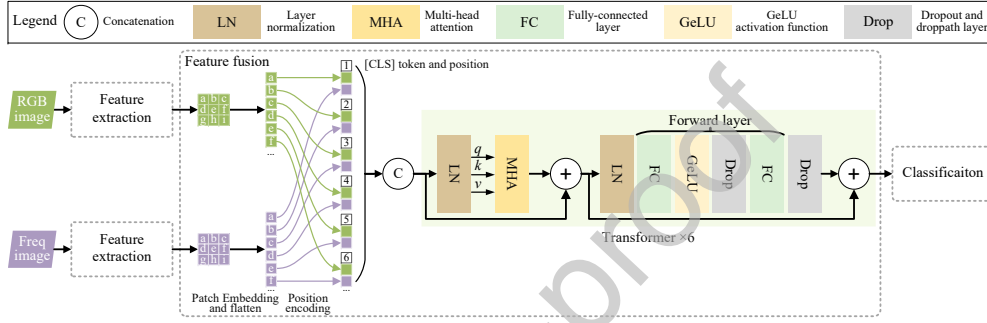


Fig. 3: The architecture of feature fusion.

236 3.4.1. Patch Embedding module

237 1) Patch Embedding

238 The spatial and frequency domain features $\tilde{\mathbf{F}}_{rgb}$ and $\tilde{\mathbf{F}}_{freq}$ are embedded
 239 by the 2D convolution layer to obtain the features.

$$\begin{aligned} \mathbf{P}_{rgb} &= \text{Conv2D}_{k=3,s=3}(\tilde{\mathbf{F}}_{rgb}), \mathbf{P}_{rgb} \in \mathbb{R}^{B \times 768 \times 2 \times 2}, \\ \mathbf{P}_{freq} &= \text{Conv2D}_{k=3,s=3}(\tilde{\mathbf{F}}_{freq}), \mathbf{P}_{freq} \in \mathbb{R}^{B \times 768 \times 2 \times 2}, \end{aligned} \quad (9)$$

240 where $\text{Conv2D}_{k=3,s=3}$ is the 2D convolution layer with kernel size of 3 and
 241 stride of 3, $\mathbf{P}_{rgb}$ and $\mathbf{P}_{freq}$ are the Patch Embedding features of spatial and
 242 frequency domain, respectively.

244 2) Patch flatten

245 After the Patch Embedding, features are flattened into the token series:

$$\begin{aligned} \mathbf{T}_{rgb} &= \text{Flatten}(\mathbf{P}_{rgb}) \in \mathbb{R}^{B \times 4 \times 768}, \\ \mathbf{T}_{freq} &= \text{Flatten}(\mathbf{P}_{freq}) \in \mathbb{R}^{B \times 4 \times 768}. \end{aligned} \quad (10)$$

246 3) Position encoding

247 Then **we add** the learnable position encoding to the token series:

$$\begin{aligned} \mathbf{T}_{rgb} &= \mathbf{T}_{rgb} + \mathbf{E}_{pos}^{rgb}, & \mathbf{E}_{pos}^{rgb} &\in \mathbb{R}^{1 \times 4 \times 768}, \\ \mathbf{T}_{freq} &= \mathbf{T}_{freq} + \mathbf{E}_{pos}^{freq}, & \mathbf{E}_{pos}^{freq} &\in \mathbb{R}^{1 \times 4 \times 768}, \end{aligned} \quad (11)$$

248 where $\mathbf{E}_{pos}^{rgb}$ and $\mathbf{E}_{pos}^{freq}$ are the learnable position encoding for spatial and
249 frequency domain, respectively.

250 4) Concatenate [CLS] token

251 **Firstly we initialize** a learnable [CLS] token $\mathbf{T}_{CLS} \in \mathbb{R}^{1 \times 1 \times 768}$ and copy
252 it to the feature batch:

$$\mathbf{T}_{CLS}^{(B)} = \text{Repeat}(\mathbf{T}_{CLS}, B), \mathbf{T}_{CLS}^{(B)} \in \mathbb{R}^{B \times 1 \times 768}. \quad (12)$$

253 Then **we concatenate the** [CLS] with the token series and obtain the
254 final token series:

$$\mathbf{T} = [\mathbf{T}_{CLS}^{(B)}; \mathbf{T}_{rgb}; \mathbf{T}_{freq}], \mathbf{T} \in \mathbb{R}^{B \times 9 \times 768}. \quad (13)$$

255 3.4.2. Transformer fusion

256 In this work, we employ a six-layer **Transformer** to capture global de-
257 pendencies and **fuse features**. **Each layer contains** the following architectures.

258 1) Multi-head attention layer

259 Firstly, **we employ** layer normalization. For input data $\mathbf{T}$, the layer
260 normalization can be expressed as:

$$\mathbf{T}_{LN} = \text{LN}(\mathbf{T}), \mathbf{T}_{LN} \in \mathbb{R}^{B \times 9 \times 768}, \quad (14)$$

261 where LN is the layer normalization function:

$$\text{LN}(x) = \gamma_L \cdot \frac{x - \mu_L}{\sqrt{\sigma_L^2 + \epsilon}} + \beta_L, \quad (15)$$

262 where μ_L and σ_L are the average and standard deviation of layer, γ_L and
263 β_L are the learnable parameters, $\epsilon = 0.00001$ is a small constant to prevent
264 division by zero.

After that, we employ the multi-head attention to capture the dependencies in the global scope. The single-head attention is calculated as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\mathbf{V}\right), \quad (16)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{B \times 9 \times 768}$ are the Query, Key and Value matrices, respectively:

$$\begin{aligned} \mathbf{Q} &= \mathbf{X}\mathbf{W}_Q, \mathbf{W}_Q \in \mathbb{R}^{768 \times 768} \\ \mathbf{K} &= \mathbf{X}\mathbf{W}_K, \mathbf{W}_K \in \mathbb{R}^{768 \times 768}, \\ \mathbf{V} &= \mathbf{X}\mathbf{W}_V, \mathbf{W}_V \in \mathbb{R}^{768 \times 768} \end{aligned} \quad (17)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are the learnable weight matrices for Query, Key and Value, respectively.

The multi-head attention (MHA) can be obtained as:

$$\begin{aligned} \text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{concat}(\text{head}_1, \dots, \text{head}_{12})\mathbf{W}_O, \\ \text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &\in \mathbb{R}^{B \times 9 \times 768}. \end{aligned} \quad (18)$$

We also employ the dropout and residual connection as follows:

$$\mathbf{T}_1 = \mathbf{T} + \text{Dropout}(\text{MHA}(\text{LN}(\mathbf{T}))), \mathbf{T}_1 \in \mathbb{R}^{B \times 9 \times 768}. \quad (19)$$

2) Feed-Forward network.

Firstly, we employ another layer normalization:

$$\mathbf{T}_2 = \text{LN}(\mathbf{T}_1). \quad (20)$$

Then we design a feed forward layer with two linear layers and a Gaussian Error Linear Unit (GeLU) activation function in between:

$$\mathbf{T}_3 = \text{FFN}(\mathbf{T}_2) = \text{Dropout}(\mathbf{W}_2 \cdot \text{Dropout}(f_G(\mathbf{W}_1 \cdot \mathbf{T}_2))), \quad (21)$$

where $f_G(x) = 0.5x \left(1 + \tanh\left(\sqrt{2/\pi}(x + 0.044715x^3)\right)\right)$ is the GeLU activation function, $\mathbf{W}_1$ and $\mathbf{W}_2$ are the learnable weight matrices.

After that, we add the residual connection:

$$\mathbf{T}_{out} = \mathbf{T}_3 + \text{Dropout}(\text{MLP}(\mathbf{T}_3)), \mathbf{T}_{out} \in \mathbb{R}^{B \times 9 \times 768}. \quad (22)$$

280 3.5. Classification

281 In this part, we employ a fully-connected layer to classify the output of the Transformer block. The architecture is shown in Fig. 4.

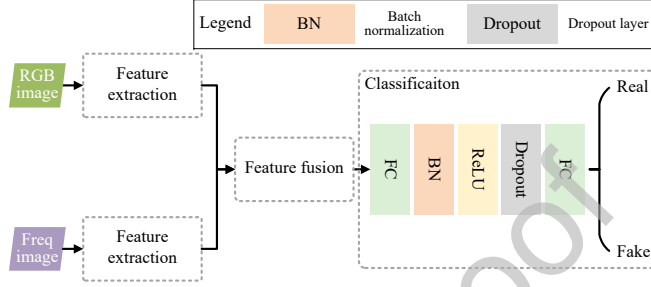


Fig. 4: The architecture of classification.

282 After the Transformer block, we obtain the output corresponding to the
283 [CLS] token:
284

$$\mathbf{v}_{cls} = \mathbf{T}_{out}[:, 0, :], \mathbf{v}_{cls} \in \mathbb{R}^{B \times 768}. \quad (23)$$

285 Then we input the $\mathbf{T}_{out}$ to the fully connected layer for classification:

$$\mathbf{h} = \sigma_R(\text{BN}(\mathbf{W}_1 \cdot \mathbf{v}_{cls})) \quad (24)$$

$$\mathbf{y} = \mathbf{W}_2 \cdot \text{Dropout}(\mathbf{h}) \in \mathbb{R}^{B \times \text{Class}} \quad (25)$$

287 where $\mathbf{W}_1$ and $\mathbf{W}_2$ are the learnable weight matrices, $\text{Class} = 2$ is the number
288 of classes, the final output $\mathbf{y}$ is the predicted probability distribution over
289 the classes, BN is the batch normalization and can be expressed as:

$$\text{BN}(x) = \gamma_B \cdot \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta_B, \quad (26)$$

290 where μ_B and σ_B are the mean and variance of the batch, respectively, γ_B
291 and β_B are the learnable parameters, $\epsilon = 0.00001$ is a small constant to
292 prevent division by zero.

4. Experiments

The experiments are shown in this Section. In Sections 4.1 and 4.2, datasets and the experimental setup will be introduced, respectively. In Sections 4.3 to 4.9, the results will be presented.

4.1. Datasets

In this work, we employ the FaceForensics++ [40] and Celeb-DF-V2 [41] datasets to evaluate the performance of the proposed method.

The FaceForensics++ includes videos manipulated by different generative model. We employ three image datasets extracted from the FaceForensics++ because our model is proposed to recognize face images: The “FaceForensics++ extracted frames” extracted from the raw FaceForensics++ and stored in png format, the “FFerop8fr” extracted from the raw FaceForensics++ but stored in jpg format, and the “Face_image8_FF++_C23” extracted from the compressed FaceForensics++ using parameter C23 and stored in png format. The dataset extracted from the raw FaceForensics++ can be treated as a standard, and then examine the adaptability of the proposed model for image or video compression. Since the FaceForensics++ is generated by different methods, we treat faces generated by each method as an independent sub-dataset, and mix all forged faces with the real faces as another sub-dataset, namely Mix. It should be noted that, the sub-datasets share same real faces, makes the number of real images in the Mix sub-dataset is not the sum of other sub-datasets, but the forged images are. Therefore, the Mix sub-dataset is significant imbalanced, which can further test the performance of models.

The Celeb-DF-V2 contains real and forged videos, we also obtain an image dataset extracted from it, namely “CelebDF_V2 (image Dataset)” stored in jpg format.

320 The basic information of datasets is shown in Table 1. For the simplicity
 321 of the paper, we only use their simplified names in the rest of paper. The
 322 detailed number of samples in each dataset is shown as Table 2.

Table 1: The basic information of datasets.

Original name	Source	Simplified name	Resolution	Division method ¹
FaceForensics++ extracted frames	FaceForensics++	FF-png	256×256	Random ²
FFcrop8fr	FaceForensics++	FF-jpg	256×256	Retain ³
Face_image8_FF++_C23	FaceForensics++	FF-C23	160×160	Retain
CelebDF_V2 (image Dataset)	Celeb-DF-V2	Celeb	256×256	Retain

1. Division of the training, validation and testing sets.
2. Randomly divide the dataset into training, validation and testing sets.
3. Retain the original division of the dataset as it uploaded.

Table 2: The number of samples in each dataset.

Dataset	Sub-Datasets	Number of samples						
		Training set		Validation set		Testing set		Total
		Real	Forged	Real	Forged	Real	Forged	
FF-png	Deepfakes	23003	22926	4472	4458	4474	4459	63792
	Face2Face	23003	23003	4472	4472	4474	4474	63898
	FaceShifter	23003	22985	4472	4469	4474	4470	63873
	FaceSwap	23003	23006	4472	4473	4474	4475	63903
	NeuralTextures	23003	22990	4472	4470	4474	4471	63880
	Mix	23003	114912	4472	22344	4474	22345	191550
FF-jpg	Deepfakes	5759	5746	1120	1111	1120	1117	15973
	Face2Face	5759	5759	1120	1120	1120	1120	15998
	FaceSwap	5759	5758	1120	1119	1120	1120	15996
	NeuralTextures	5759	5753	1120	1120	1120	1120	15992
	Mix	5759	23016	1120	4470	1120	4477	39962
FF-C23	Deepfakes	5367	5384	771	780	1529	1542	15373
	Face2Face	5367	5438	771	773	1529	1551	15429
	FaceShifter	5367	5440	771	776	1529	1550	15433
	FaceSwap	5367	5431	771	772	1529	1549	15419
	NeuralTextures	5367	5432	771	780	1529	1546	15425
	Mix	5367	27125	771	3881	1529	7738	46411
Celeb V2	N/A	40288	40536	5036	5068	5036	5067	101031

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4.2. Experimental setup

Image preprocessing: The data preprocessing method for the training, validation and testing sets are shown in Table 3. The sample of original, preprocessed and DCT-transformed faces are shown in Fig. 5.

Table 3: The image preprocessing method.

Stage	Method	Value
Training	Scale and clip	FF-C23: scale images to 256×256 , then randomly clip to 224×224 . Others: randomly clip to 224×224 .
	Random rotate	Max rotation degree = 45° .
	Random vertical flip	Flip probability = 0.5.
	Random horizontal flip	Flip probability = 0.5.
	Color jitter	Brightness=0.3, Contrast=0.2, Saturation=0.2, Hue=0.3.
	z-score normalize	Means = [0.485, 0.456, 0.406], standard deviations = [0.229, 0.224, 0.225] (pretrained from ImageNet).
Validation and testing	Scale and clip	FF-C23: scale images to 256×256 , then center clip to 224×224 . Others: only center clip to 224×224 .
	z-score normalize	Means = [0.485, 0.456, 0.406], standard deviations = [0.229, 0.224, 0.225] (pretrained from ImageNet).

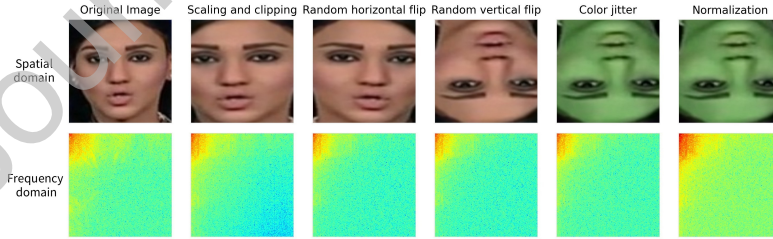


Fig. 5: The spatial and frequency images samples.

Hyperparameters: The training batch size is set to 32, and the max training epochs is 25. The loss function is the Focal loss [42], the optimizer is

AdamW algorithm [43], which introduces the weight attenuation decoupling mechanism on the traditional Adam. The learning rate is updated as:

$$lr = \eta \gamma^{\lfloor \frac{epoch}{step} \rfloor}, \quad (27)$$

where $\eta = 0.0001$ is the initial learning rate, $\gamma = 0.1$ is the decay factor, and $step = 7$ is the number of epochs for each decay.

Comparison algorithms: We compare the proposed SFD-ViT with three baseline networks, namely ResNet50 (RN, [44]), ENV2S (EN, [39]) and Swin Transformer (ST, [45]). We also compare three **State-of-the-Art** (SOTA) methods, namely **DNA-Det** [22], FreqNet [29] and GocNet [25].

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Evaluation metrics: In this work, we set **the forged faces** as positive samples. Hence, the true positive (TP) is the recognized forged faces, the false negative (FN) is the misdetected forged faces, the false positive (FP) is the misdetected real faces, and the true negative (TN) is the recognized real faces. We examined the performance of the model by five evaluation metrics, including accuracy, precision, recall, F1-Score, and Area Under Curve (AUC). These metrics can be obtained as following formulae.

$$\begin{aligned} accuracy &= \frac{TP+TN}{TP+TN+FP+FN} \\ precision &= \frac{TP}{TP+FP} \\ recall &= \frac{TP}{TP+FN} \\ F1 - Score &= 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \\ AUC &= \frac{sum_{positive} - \frac{M(M+1)}{2}}{M \times N} \end{aligned}, \quad (28)$$

where m and n are the number of positive and negative samples, respectively, $sum_{positive}$ is the sum of the rankings of all positive samples.

Hardware: The deep learning frameworks in this work are PyTorch 2.8.0 and CUDA 12.9. The training is accelerated by Nvidia[®] GeForce RTX-5080 with 16GB VRAM and RTX-4090 with 24GB VRAM.

4.3. Results for FF-png

The results of the FF-png dataset are shown in Table 4. The **bold** and underscore values indicate the best and the second-best indices, respectively. It should be noted that, to avoid awarding the optimal value to models that achieve high scores on metrics by extreme prediction (predicting all samples as the same category), given that F1-Score and AUC are more reliable indicators for measuring the overall performance, we only label them for a more fair comparison in the Mix sub-dataset due to its severe class imbalance. In the subsequent presentation of results involving imbalanced datasets in this paper, we will follow this rule and will not elaborate further.

The results indicate that, with regard to the accuracy, the SFD-ViT achieves the highest in all sub-datasets. With regard to the precision and recall, the situation is different, the proposed method is inferior to other methods in some datasets. Nevertheless, given that face forgery detection is a complex problem and the Mix is imbalanced, it is necessary to select indicators that can provide a global evaluation of performance. The F1-Score indicate that our model achieves the best in all cases, and the AUC indicate that our model also outperforms in most cases. With regard to the comparing methods, although there are some fluctuations, all models except for DNA-Det and FreqNet perform well. The DNA-Det performs poor in some sub-datasets (such as NeuralTextures), indicating that it is sensitive to certain image forgery methods. However, the FreqNet performs worse than other methods in all models. The results also indicate that there are differences in the difficulty of recognizing forged faces generated by different methods. In conclusion, the comprehensive performance of the proposed SFD-ViT is the best among all the compared models.

Table 4: The testing metrics of the FF-png dataset.

Sub-Datasets	Model	Accuracy	Precision	Recall	F1Score	AUC
Deepfakes	RN	98.825%	99.840%	97.802%	0.98810	0.99926
	EN	99.586%	99.865%	99.305%	0.99584	0.99997
	ST	99.687%	99.843%	99.529%	0.99686	0.99983
	DNA-Det	94.962%	98.950%	90.872%	0.94739	0.99506
	FreqNet	84.999%	90.370%	78.291%	0.83898	0.94029
	GocNet	98.802%	99.863%	97.735%	0.98787	0.99928
	SFD-ViT	99.720%	99.887%	99.551%	0.99719	0.99998
Face2Face	RN	98.815%	99.682%	97.944%	0.98805	0.99853
	EN	99.620%	99.377%	99.866%	0.99621	0.99997
	ST	99.061%	99.909%	98.212%	0.99053	0.99990
	DNA-Det	70.988%	71.566%	69.647%	0.70594	0.78156
	FreqNet	82.477%	86.526%	76.933%	0.81448	0.91095
	GocNet	98.458%	99.050%	97.854%	0.98448	0.99892
	SFD-ViT	99.676%	99.821%	99.531%	0.99675	0.99998
FaceShifter	RN	96.109%	99.639%	92.550%	0.95964	0.99866
	EN	99.396%	99.374%	99.418%	0.99396	0.99986
	ST	99.486%	99.820%	99.150%	0.99484	0.99975
	DNA-Det	97.585%	98.673%	96.465%	0.97557	0.99759
	FreqNet	80.221%	79.766%	80.962%	0.80360	0.88668
	GocNet	98.960%	99.615%	98.300%	0.98953	0.99913
	SFD-ViT	99.564%	99.955%	99.172%	0.99562	0.99992
FaceSwap	RN	98.883%	98.428%	99.352%	0.98888	0.99953
	EN	99.732%	99.621%	99.844%	0.99732	0.99995
	ST	99.575%	99.355%	99.799%	0.99576	0.99996
	DNA-Det	67.281%	65.332%	73.654%	0.69244	0.74279
	FreqNet	75.964%	76.421%	75.106%	0.75758	0.84610
	GocNet	97.396%	96.129%	98.771%	0.97432	0.99733
	SFD-ViT	99.866%	99.888%	99.844%	0.99866	0.99999
NeuralTextures	RN	90.129%	98.122%	81.816%	0.89230	0.97967
	EN	97.328%	99.625%	95.012%	0.97264	0.99889
	ST	95.942%	99.542%	92.306%	0.95787	0.99788
	DNA-Det	60.157%	64.758%	44.509%	0.52757	0.65088
	FreqNet	66.831%	75.526%	49.765%	0.59997	0.74376
	GocNet	90.665%	98.223%	82.823%	0.89868	0.97970
	SFD-ViT	98.245%	99.382%	97.092%	0.98224	0.99933
Mix	RN	91.372%	97.406%	92.097%	0.94677	0.96499
	EN	97.763%	99.762%	97.548%	0.98642	0.99868
	ST	97.789%	99.102%	98.237%	0.98667	0.99641
	DNA-Det	83.281%	84.276%	98.268%	0.90736	0.70044
	FreqNet	83.228%	83.375%	99.763%	0.90836	0.68724
	GocNet	91.812%	98.155%	91.900%	0.94924	0.97396
	SFD-ViT	98.016%	99.659%	97.955%	0.98799	0.99844

376 4.4. Results for FF-jpg

377 The results of the FF-jpg dataset are shown in Table 5. The bold and
 378 underscore values indicate the best and the second-best indices, respectively.

Table 5: The testing metrics of the FF-jpg dataset.

Sub-Datasets	Model	Accuracy	Precision	Recall	F1Score	AUC
Deepfakes	RN	93.608%	90.516%	97.404%	0.93834	0.98908
	EN	96.603%	98.599%	94.539%	0.96527	0.99638
	ST	96.871%	97.120%	96.598%	0.96858	0.99670
	DNA-Det	87.215%	87.399%	86.929%	0.87163	0.95029
	FreqNet	81.851%	79.551%	85.676%	0.82500	0.90320
	GocNet	94.904%	95.140%	94.628%	0.94883	0.99062
	SFD-ViT	97.720%	98.810%	96.598%	0.97691	0.99759
Face2Face	RN	94.196%	95.082%	93.214%	0.94139	0.98334
	EN	<u>97.857%</u>	98.029%	97.679%	<u>0.97853</u>	0.99093
	ST	<u>97.857%</u>	<u>98.463%</u>	<u>97.232%</u>	0.97844	<u>0.99123</u>
	DNA-Det	65.446%	62.482%	77.321%	0.69114	0.71311
	FreqNet	67.545%	86.188%	41.786%	0.56284	0.78252
	GocNet	94.643%	97.619%	91.518%	0.94470	0.98330
	SFD-ViT	98.036%	98.820%	<u>97.232%</u>	0.98020	0.99177
FaceSwap	RN	95.848%	96.809%	94.821%	0.95805	0.99197
	EN	<u>97.232%</u>	95.920%	98.661%	<u>0.97271</u>	0.99794
	ST	97.054%	95.826%	<u>98.393%</u>	0.97093	<u>0.99765</u>
	DNA-Det	64.821%	62.351%	74.821%	0.68019	0.70818
	FreqNet	69.643%	80.387%	51.964%	0.63124	0.81293
	GocNet	93.616%	93.040%	94.286%	0.93659	0.98163
	SFD-ViT	97.589%	97.336%	97.857%	0.97596	0.99746
NeuralTextures	RN	83.750%	89.457%	76.518%	0.82483	0.92031
	EN	84.420%	<u>97.070%</u>	70.982%	0.82001	0.95343
	ST	<u>88.482%</u>	97.783%	<u>78.750%</u>	<u>0.87240</u>	0.96989
	DNA-Det	59.821%	59.354%	62.321%	0.60801	0.64252
	FreqNet	58.170%	69.848%	28.750%	0.40734	0.67451
	GocNet	83.705%	89.612%	76.250%	0.82393	0.92138
	SFD-ViT	90.268%	95.648%	84.375%	0.89658	<u>0.96200</u>
Mix	RN	81.311%	82.107%	97.990%	0.89348	0.74250
	EN	93.622%	98.108%	93.835%	<u>0.95924</u>	<u>0.98064</u>
	ST	91.406%	93.435%	96.002%	0.94701	0.96242
	DNA-Det	80.114%	80.548%	99.062%	0.88851	0.65383
	FreqNet	79.918%	80.532%	98.771%	0.88724	0.66722
	GocNet	84.116%	88.564%	92.026%	0.90262	0.86476
	SFD-ViT	94.318%	97.144%	95.711%	0.96422	0.98183

379 The metrics indicate that the JPG-format images decrease the perfor-

mance of models, manifested as decreases in all metrics. Nevertheless, the proposed model continues to demonstrate superior performance in terms of accuracy, particularly in the NeuralTextures sub-dataset where it is the only model to achieve an accuracy exceeding 90%. Despite exhibiting lower precision and recall metrics in certain instances when compared to other models, our model obtains the best AUC in three sub-datasets, which means that the model is slightly inferior at the probability level, but the F1-Score of our model is best in all cases, indicating that it is more stable in making correct decisions. [The performance trends of different methods are similar to that of the FF-png, and the DNA-Det and FreqNet having accuracies below 60% in the NeuralTextures sub-dataset.](#) In conclusion, the comprehensive performance of the proposed SFD-ViT remains the best.

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4.5. Results for FF-C23

The results of the FF-C23 dataset are shown in [Table 6](#). [The bold and underscore values indicate the best and the second-best indices, respectively.](#)

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The results of FF-C23 are similar to that of the FF-png, where our model achieves the best accuracy and F1-Score. [Though the AUC ranks second in some cases, the high accuracy and F1-Score indicating its stronger generalization and robustness under video compression.](#) Although the AUC is slightly lower, the overall [performance](#) is consistent, indicating that the model is more reliable in practical [application](#).

4.6. Results for Celeb

The results of the Celeb dataset are shown in [Table 7](#). [The bold and underscore values indicate the best and the second-best indices, respectively.](#)

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The testing metrics indicate that almost all of the models (except FreqNet) perform well on the Celeb dataset, indicating that the recognition difficulty of this dataset is relatively low. However, the proposed model still outperforms other models in all metrics.

Table 6: The testing metrics of the FF-C23 dataset.

Sub-Datasets	Model	Accuracy	Precision	Recall	F1Score	AUC
Deepfakes	RN	92.999%	96.890%	88.911%	0.92729	0.98322
	EN	94.106%	97.754%	90.337%	0.93900	0.98633
	ST	<u>95.344%</u>	93.366%	97.665%	<u>0.95468</u>	0.99269
	DNA-Det	85.510%	89.319%	80.804%	0.84848	0.92955
	FreqNet	65.125%	91.979%	33.463%	0.49073	0.81021
	GocNet	93.097%	95.300%	90.726%	0.92957	0.98329
	SFD-ViT	95.962%	98.562%	<u>93.320%</u>	0.95869	<u>0.99020</u>
Face2Face	RN	84.935%	79.080%	95.293%	0.86433	0.94999
	EN	<u>94.383%</u>	<u>94.110%</u>	94.778%	<u>0.94443</u>	0.97918
	ST	94.156%	93.086%	95.487%	<u>0.94271</u>	<u>0.98631</u>
	DNA-Det	59.318%	59.430%	60.542%	<u>0.59981</u>	0.62705
	FreqNet	56.558%	65.193%	29.465%	0.40586	0.62416
	GocNet	90.909%	91.892%	89.877%	0.90874	0.96309
	SFD-ViT	95.552%	95.731%	<u>95.422%</u>	0.95576	0.98698
FaceShifter	RN	90.224%	97.418%	82.774%	0.89501	0.98596
	EN	<u>96.038%</u>	<u>97.664%</u>	94.387%	<u>0.95997</u>	0.99460
	ST	94.868%	99.014%	90.710%	0.94680	0.99687
	DNA-Det	90.711%	92.021%	89.290%	0.90635	0.96861
	FreqNet	62.358%	86.271%	30.000%	0.44519	0.79984
	GocNet	<u>94.284%</u>	<u>95.497%</u>	93.032%	0.94248	0.98737
	SFD-ViT	96.330%	<u>98.449%</u>	<u>94.194%</u>	0.96274	<u>0.99638</u>
FaceSwap	RN	92.560%	89.333%	96.772%	0.92904	0.97899
	EN	94.997%	91.993%	98.644%	0.95202	0.99352
	ST	<u>96.751%</u>	98.268%	95.223%	<u>0.96721</u>	<u>0.99366</u>
	DNA-Det	67.836%	66.747%	71.917%	0.69236	0.73604
	FreqNet	58.317%	56.051%	79.535%	0.65759	0.63638
	GocNet	89.929%	87.252%	93.673%	0.90349	0.96771
	SFD-ViT	97.336%	<u>97.110%</u>	<u>97.611%</u>	0.97360	0.99496
NeuralTextures	RN	65.821%	75.542%	47.348%	0.58211	0.74823
	EN	74.634%	80.157%	65.847%	0.72301	0.83822
	ST	<u>78.374%</u>	<u>82.224%</u>	72.704%	<u>0.77171</u>	<u>0.86185</u>
	DNA-Det	56.553%	58.190%	48.254%	0.52758	0.59051
	FreqNet	53.886%	67.204%	16.171%	0.26069	0.63963
	GocNet	73.073%	81.109%	60.543%	0.69333	0.81569
	SFD-ViT	80.911%	88.514%	<u>71.281%</u>	0.78968	0.89923
Mix	RN	82.055%	86.139%	93.564%	0.89698	0.68151
	EN	87.472%	91.537%	93.655%	<u>0.92584</u>	<u>0.90033</u>
	ST	85.238%	93.204%	88.796%	0.90946	0.88938
	DNA-Det	83.587%	84.081%	99.108%	0.90978	0.65833
	FreqNet	83.501%	83.501%	100.000%	0.91009	0.61304
	GocNet	83.900%	85.643%	96.976%	0.90958	0.76388
	SFD-ViT	88.249%	94.344%	91.406%	0.92852	0.92446

Table 7: The testing metrics of the Celeb dataset.

Model	Accuracy	Precision	Recall	F1Score	AUC
RN	98.882%	98.493%	99.290%	0.98889	0.99945
EN	<u>99.782%</u>	<u>99.724%</u>	<u>99.842%</u>	<u>0.99783</u>	<u>0.99998</u>
ST	99.723%	99.704%	99.743%	0.99724	0.99997
DNA-Det	94.437%	95.349%	93.468%	0.94399	0.98698
FreqNet	64.347%	68.326%	53.898%	0.60260	0.72889
GocNet	98.169%	98.918%	97.415%	0.98160	0.99852
SFD-ViT	99.871%	99.862%	99.882%	0.99872	0.99999

4.7. Cross-dataset testing

To fully test the generalization of models, we conduct **cross-dataset** experiments. Specifically, models trained on the Mix sub-dataset in FF-png, FF-jpg and FF-C23 datasets are used to recognize the images in the Celeb dataset and obtain evaluation metrics. The results are shown in Table 8. The **bold** and underscore values indicate the best and the second-best indices, respectively. The optimal value is only annotated on the F1-Score and AUC, because many models make extreme predictions that fail to accurately reflect their performance in terms of the accuracy, precision, and recall.

The results indicate that all models **suffer** a significant performance decline in **cross-dataset** recognition. For the FF-png/Mix dataset, the DNA-Det and FreqNet achieve accuracy of around 50%, indicating that they are **unable to classify**. The cross-dataset performance of the models decrease with the dataset compression, indicating that the image characteristics of FF-png dataset are similar to those of the Celeb, and dataset compression leads to the disappearance of this similarity. The F1-Score and AUC indicate that the cross-dataset recognition ability of proposed model is inferior to the ST model, ranking second. Overall, the **cross-dataset** performance of all models is unsatisfying, indicates that there is still a lot of room for improvement.

Table 8: The cross-dataset testing results.

Source dataset of model	Model	Accuracy	Precision	Recall	F1Score	AUC
FF-png/Mix	RN	61.358%	58.868%	76.179%	0.66414	0.66669
	EN	66.485%	66.471%	66.943%	0.66706	0.73272
	ST	72.503%	67.908%	85.652%	0.75755	0.83645
	DNA-Det	50.292%	50.223%	100.000%	0.66865	0.63820
	FreqNet	50.332%	50.247%	98.224%	0.66484	0.57553
	GocNet	63.070%	70.745%	44.958%	0.54978	0.70054
	SFD-ViT	67.406%	63.812%	80.876%	<u>0.71338</u>	<u>0.76119</u>
FF-jpg/Mix	RN	50.747%	50.458%	98.875%	0.66818	0.54993
	EN	65.149%	64.563%	67.634%	0.66063	0.72276
	ST	62.061%	57.288%	95.717%	0.71677	0.77620
	DNA-Det	50.737%	50.451%	99.309%	0.66910	0.57368
	FreqNet	51.460%	50.929%	88.198%	0.64572	0.56723
	GocNet	58.062%	58.293%	57.569%	0.57929	0.60617
	SFD-ViT	67.673%	65.626%	74.640%	<u>0.69843</u>	<u>0.74486</u>
FF-C23/Mix	RN	50.094%	50.124%	99.428%	0.66649	0.57547
	EN	54.429%	52.492%	96.231%	0.67930	0.60497
	ST	60.022%	56.132%	92.856%	0.69968	0.72319
	DNA-Det	50.619%	50.392%	99.033%	0.66795	0.56825
	FreqNet	50.153%	50.153%	100.000%	0.66803	0.45208
	GocNet	50.846%	50.508%	99.092%	0.66911	0.57498
	SFD-ViT	56.953%	54.080%	93.921%	<u>0.68638</u>	<u>0.67846</u>

4.8. Robustness testing

To verify the robustness of the model, we apply interferences to the **test-**
ing set of the Mix sub-dataset in the FF-C23 dataset, including the Gaussian
blur with $kernel = (3, 3)$ and Gaussian noise with $\sigma = 5$, **thereby generat-**
ing the interfered dataset. Subsequently, we use the corresponding model to
recognize **the interfered dataset** to obtain evaluation metrics. The results are
shown in Table 9. **The bold** and **underscore** values indicate the best and the
second-best indices, respectively.

The results show that the performance of all models except RN and
FreqNet decreased compare to the standard (**uninterfered image**). The SFD-
ViT and EN, which originally performed well, showing the largest decrease.
Overall, while facing the interference of blur and noise, our model still main-

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Table 9: The robustness testing results.

Interference	Model	Accuracy	Precision	Recall	F1Score	AUC
Standard	RN	82.055%	86.139%	93.564%	0.89698	0.68151
	EN	87.472%	91.537%	93.655%	<u>0.92584</u>	<u>0.90033</u>
	ST	85.238%	93.204%	88.796%	0.90946	0.88938
	DNA-Det	83.587%	84.081%	99.108%	0.90978	0.65833
	FreqNet	83.501%	83.501%	100.000%	0.91009	0.61304
	GocNet	83.900%	85.643%	96.976%	0.90958	0.76388
	SFD-ViT	88.249%	94.344%	91.406%	0.92852	0.92446
Blur	RN	83.630%	84.088%	99.160%	0.91004	0.66635
	EN	84.450%	84.871%	99.031%	<u>0.91406</u>	<u>0.79708</u>
	ST	84.407%	85.833%	97.402%	0.91252	0.78086
	DNA-Det	83.468%	83.517%	99.922%	0.90986	0.63381
	FreqNet	83.501%	83.501%	100.000%	0.91009	0.64505
	GocNet	83.781%	84.643%	98.436%	0.91020	0.72799
	SFD	84.526%	84.607%	99.586%	0.91488	0.85622
Noise	RN	81.914%	86.032%	93.525%	0.89622	0.67961
	EN	83.673%	83.652%	99.987%	<u>0.91093</u>	0.64732
	ST	75.785%	92.171%	77.591%	0.84255	0.79653
	DNA-Det	83.630%	83.969%	99.367%	0.91021	0.65944
	FreqNet	83.501%	83.501%	100.000%	0.91009	0.47762
	GocNet	83.501%	83.501%	100.000%	0.91009	0.58062
	SFD	84.634%	85.062%	98.979%	0.91494	<u>0.75046</u>

tains the best F1-Score and AUC, indicating that the proposed model still has advantages. However, the overall performance decline indicates that the robustness still needs to be improved.

4.9. Computational efficiency

To provide a comprehensive evaluation of the computational efficiency of the proposed SFD-ViT and other competing models, we measure the training and validating speed by training one epoch of each model. The aforementioned measurements are conducted on the Deepfakes sub-dataset in FF-C23 dataset. All experiments are performed on a standardized hardware configuration (Nvidia[®] GeForce RTX-5080 with 16GB VRAM) and hyperparameters to ensure a fair comparison. The results are shown in Table 10.

The results indicate that the DNA-Det achieves fastest training and

R4Q1

Table 10: The computational efficiency comparison.

Model	Training speed (iters/s)	Validating speed (images/s)
RN	20.19	1678.08
EN	17.14	1771.20
ST	6.06	543.68
DNA-Det	36.55	2508.16
FreqNet	3.28	270.08
GocNet	25.01	1928.00
SFD-ViT	7.78	856.00

validating speeds, followed by the GocNet, indicating that these two SOTAs are more efficient. The proposed SFD-ViT has a complex structure, making the computational efficiency inferior to most models. However, the speed of the SFD-ViT is still faster than ST and FreqNet, which is still acceptable, and it has practical value in situations where the highest accuracy is required.

5. Ablation study

In this section, we design ablation experiments to verify the effectiveness of each module in the proposed architecture. The ablation study setup is described in Section 5.1, and the results are presented in Section 5.2.

5.1. Setup

The model architecture proposed in this work (shown in Fig. 1) consists of four modules: spatial-domain feature extraction, frequency-domain feature extraction, feature fusion and classification. Hence, we design the ablation study containing six situations, shown as follows.

Situation 0 (S0): the proposed model, including all modules.

S1: remove the spatial-domain feature extraction, retain others.

S2: remove the frequency-domain feature extraction, retain others.

S3: remove the SE in the spatial and frequency-domain feature extraction module, retain others.

470 **S4**: replace whole feature fusion block (including the Patch Embedding
 471 and Transformer) with the addition, **retain others**.

472 **S5**: replace the Patch Embedding with the addition, **retain others**.

473 **S6**: replace the Transformer with the addition, **retain others**.

The structure of each situation is shown in Fig. 6.

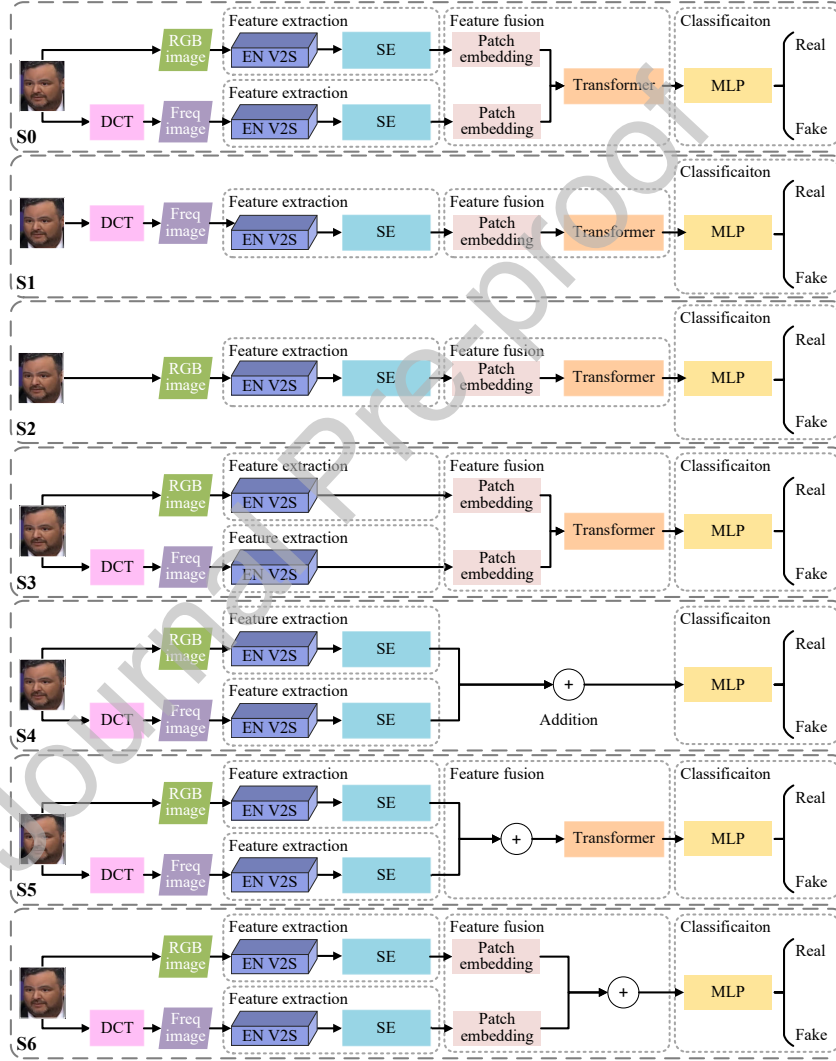


Fig. 6: The ablation experiments setup.

474

5.2. Results

We perform the ablation experiments on the Mix sub-dataset of FF-jpg dataset and the Celeb dataset, the results are shown in Table 11. The **bold** values indicate the best index.

Table 11: **The ablation study results.**

Dataset	Model	Accuracy	Precision	Recall	F1Score	AUC
FF-jpg/Mix	S0	94.318%	97.144%	95.711%	0.96422	0.98183
	S1	79.989%	79.989%	100.000%	0.88882	0.50798
	S2	94.068%	96.189%	96.404%	0.96296	0.97853
	S3	93.854%	96.616%	95.667%	0.96139	0.97727
	S4	93.604%	96.584%	95.376%	0.95977	0.97377
	S5	94.211%	96.402%	96.359%	0.96381	0.97930
	S6	94.175%	97.138%	95.533%	0.96329	0.98061
Celeb	S0	99.871%	99.862%	99.882%	0.99872	0.99999
	S1	83.351%	79.636%	89.757%	0.84394	0.92320
	S2	99.842%	99.941%	99.743%	0.99842	0.99999
	S3	99.822%	99.901%	99.743%	0.99822	0.99998
	S4	99.822%	99.941%	99.704%	0.99822	0.99997
	S5	99.753%	99.724%	99.783%	0.99753	0.99996
	S6	99.822%	99.881%	99.763%	0.99822	0.99999

The ablation study verifies the necessity and effectiveness of each core module in the proposed SFD-ViT. The results indicate that, the proposed method achieves competitive performance, validating the effectiveness of each component. The significance of different situations is shown as follows.

The effectiveness of the dual-stream structure (S1 and S2): When only using frequency domain streams (S1) or spatial streams (S2), the model shows a decline in almost all metrics. The S1 results in significant performance degradation, which demonstrates the critical importance of **visible facial texture details**. The S2 impacts less than S1, but still confirming that frequency-domain features can capture forgery artifacts and distortions. The forged faces often introduce traces in both spatial domain (such as unnatural texture) and frequency domain (such as abnormal spectral statistical characteristics) during the generation process, making the single domain cannot capture all traces. For example, some advanced forgery methods are difficult

R4Q2

to distinguish with the naked eye in the spatial domain, but their frequency domain statistical characteristics can expose anomalies. Therefore, removing any of them would result in the loss of some key discriminative features, leading to a decline in model performance.

The role of the SE module (S3): The SE module dynamically calibrates features, allowing the network to pay more attention to the feature channels that are most effective for classification. Without the SE module for channel attention weighting, the model treats all feature channels equally, and redundant or noisy channels may interfere with the final decision. Therefore, removing it means that the model loses its adaptive feature refinement ability, thereby affecting the quality of features before input the fusion block.

R4Q2

The superiority of the feature fusion block (S4, S5, S6): The performance loss of replacing the entire fusion block with addition (S4) is significant, proving that the Transformer with Patch Embedding surpasses simple feature addition. The results of replacing only Patch Embedding (S5) and replacing only Transformer (S6) are similar but better than S4, indicating that using these two modules alone improves model performance, but still inferior to using them simultaneously. Simple addition fusion is a “shallow” operation that assumes the dual-stream features can be directly aligned and overlaid. However, the spatial and frequency domain features are extracted from different representation spaces, and there are complex and nonlinear relationships between them. Addition cannot fully model this relationship, which may result in insufficient information fusion or even conflicts. The fusion block transforms features into serialized semantic blocks, and then use self-attention mechanism to mine global associations between blocks, thus achieving a “deeper” and context-based understanding and fusion.

Regarding the performance differences on different datasets, the FF-jpg/Mix dataset has a larger difference while that of the Celeb is small. At the same time, the recall of S0 is slightly weaker in the FF-jpg/Mix dataset

while precision is slightly weaker on Celeb. Nevertheless, the proposed model (S0) maintains a leading position in comprehensive evaluation metrics such as F1-score and AUC, demonstrating its strong performance.

It should be noted that the recall of S1 of FF-jpg/Mix is 100%, but the reason is the model predicts all the samples as forged faces (accuracy = 79.99%, while the proportion of forged faces is just about 80%). Hence, the model is unusable, and it cannot be treated as the best recall. At the same time, we also notice that the AUC index of all situations except for the S1 are near to 1, which once again indicates that the recognition difficulty of the Celeb dataset is relatively low. Therefore, more attention should be paid to the accuracy and F1-Score.

6. Summary and Discussion

6.1. Summary

Precision and recall: We acknowledge that for some datasets, other models may achieve marginally higher precision or recall. On the one hand, these models might exhibit higher specialization on the specific face forgery method. In contrast, our approach is designed to learn more universal and fundamental forgery traces, which may not be the best on a single dataset or indicator but deliver consistently high performance across the board. On another hand, the F1-Score provides a more balanced view than the precision or recall. It can be observed that our method achieves the best F1-Score on all datasets, which indicates that our method strikes a superior balance between avoiding false positives (precision) and capturing all forgeries (recall).

NeuralTextures sub-dataset: Matrices of the NeuralTextures sub-dataset are inferior to others across all models. The NeuralTextures generative model is not just simply replacing the facial texture, but enabling precise and consistent control of facial dynamics by learning neural texture

map. Therefore, we believe that the performance of all models is reduced because the face generated by this method are more realistic.

Generalization ability: The generalization performance of the model should be manifested in two aspects. On the one hand, the model performs well on FaceForensics++ and non-FaceForensics++ datasets, the accuracy and F1-Score are the best, while the AUC also ranks first or second. However, on the other hand, the results of cross and interfered datasets recognition are unsatisfying. Although the SFD-ViT has the highest accuracy when trained by the FF-jpg/Mix dataset and tested by the Celeb dataset, but the F1-Score and AUC are inferior to the ST model and ranking second in all situations. At the same time, all performance indicators of all models are low, indicating that further improvement is still needed.

R4Q1

Data compression: The compression of videos or images impact on the performance of the model. All models perform optimally on the FF-png, followed by the FF-jpg, while the FF-C23 performs the worst, but our model remains the best in different compression situations. Currently, the advent of mobile Internet has precipitated a significant increase in the use of video and image compression. Consequently, the development of face forgery detection models should be improved in recognizing compressed images or videos.

Data imbalance: We construct evaluations incorporating both balanced and imbalanced datasets. Consequently, the F1-Score and AUC were selected as the core evaluation indicators, given their ability to withstand category imbalances. The proposed SFD-ViT model provides the higher F1-Score and the AUC on the Mix sub-datasets, demonstrating that our model is capable of providing enhanced resistance to performance degradation caused by data distribution bias.

Ablation study: The ablation study confirms that, the absence of the spatial or frequency stream weakens the detection of forgery artifacts; removing the SE module making the model reduces focus on important features,

replacing the whole feature fusion block or one of the components **also affect the fusion of the multi-domain features**. These results prove the rationality of the model design.

Limitations: Though the SFD-ViT model performs well on face forgery detection, some limitations still need to be pointed out. Firstly, the model has a more complex structure than competing models. **Thereby**, more hardware resources for training and testing are required, **constituting a limitation for high-speed recognition scenarios**. Secondly, although the model outperforms other models in both FF-jpg and FF-C23, further improvement is required to enhance its performance under data compression. In addition, similar to other compared models, our model still has a lot of room for improvement in cross and interfered datasets recognition. Finally, the precision and recall metrics of the proposed model performed slightly poor, indicating that the further improvement is needed in the application of the model.

Performance of various models: There are significant differences in the performance of different detectors. The RN **demonstrates poor performance** and it can not meet the requirement for complex face forgery detection. The EN and ST models outperform the proposed model in some cases, but they are inferior to the proposed method in terms of the F1-Score, which offer a comprehensive assessment of performance. The DNA-Det works well in some sub-datasets, but it is quite sensitive to forgery generation method. The FreqNet demonstrates poor performance, indicates that using only frequency-domain characteristics cannot lead in the experiments of this work. The GocNet performs superior to our model in AUC for some datasets, but it is inferior to ours in accuracy and F1-Score. In conclusion, the proposed SFD-ViT method demonstrates superior overall performance.

R4Q1

604 6.2. Discussion

605 In this work, a novel SFD-ViT is proposed for detecting multiple types
 606 of face forgery. The core innovation of the SFD-ViT lies in the spatial and
 607 frequency-domain information, as well as the employment of SE module and
 608 Transformer with Patch Embedding. The results indicate that the SFD-ViT
 609 demonstrates the best accuracy and F1-Score, and exhibiting strong perfor-
 610 mance in the AUC. The ablation study also shows that the performance of
 611 the model is diminished by the removal of key components, thereby substan-
 612 tiating the effectiveness and necessity of the proposed architecture. The roles
 613 of each key module of the SFD-ViT are discussed in detail as follows.

614 **Spatial-frequency dual-stream feature:** The dual-stream structure
 615 provides complementary and robust information sources in both the spatial
 616 and frequency domain. Experimental results demonstrate that incorporating
 617 dual-stream feature helping the model to outperform methods that are solely
 618 dependent on single domains.

619 **ENV2S feature extractor:** The ENV2S transforms the dual-stream
 620 input into the high-dimensional and informative feature maps. It exhibits
 621 superior computational efficiency and parameter utilization in comparison to
 622 the standard CNN, which is important for handling multi-domain inputs.

623 **SE module:** In the face forgery detection task, the forgery traces may
 624 be distributed across regions or feature channels. The SE module automati-
 625 cally focus on the most discriminative features, which enhances the sensitivity
 626 in capturing weak forgery traces from multi-domain features.

627 **Patch Embedding module:** The Patch Embedding module splits fea-
 628 tures into fixed-size non-overlapping image patches and maps them with the
 629 positional information to an embedding vector. Thereby, it provides the basis
 630 for the Transformer to learn global relationships.

631 **Transformer Module:** The Transformer efficiently establishes global
 632 relationships between feature channels by the self-attention mechanism, then

integrates information from the entire **feature map** and identifies global forged artifacts.

In summary, the SFD-ViT effectively integrates complementary information through spatial-frequency dual-stream inputs. It utilizes ENV2S to achieve efficient feature extraction and introduces the SE module for adaptive feature calibration to enhance discriminative properties. Furthermore, the Transformer **with Patch Embedding** are adapted to capture long-range dependencies of forgery **traces**. The SFD-ViT exhibits excellent performance, demonstrates the importance of fusing multi-domain information for face forgery detection.

7. Conclusion

The security and reliability of face forgery detection technology is directly related to personal privacy and social trust. Hence, there is an urgent need to develop detection techniques to deal with the face forgery produced by advanced generative models. To address this challenge, this work proposes the Spatial-Frequency Dual-Stream Vision Transformer (SFD-ViT) **detecting** method. The SFD-ViT utilizes both the original RGB image and its frequency-domain image (obtained by the Discrete Cosine Transform) as inputs to perceive weak forged traces. It adopts the lightweight EfficientNetV2-S for feature extraction and integrates the Squeeze-and-Excitation module for feature channel recalibration. Subsequently, the feature maps are converted into serialized tokens by Patch Embedding and input tokens into the Transformer module. The prediction results are finally obtained by the fully-connected layers. The experiments and ablation study indicate that the SFD-ViT demonstrates excellent detecting performance. In future research, the focus will be on the development of more sophisticated and nuanced representations to enhance the performance of cross or **interfered** datasets.

660 Additionally, the model architecture will be optimized to align with the low-
661 latency and high-efficiency demands of real-world application scenarios.

662 **CRediT authorship contribution statement**

663 **Yikang Yang:** Validation, Formal analysis, Investigation, Resources,
664 Writing - Original Draft, Writing - Review & Editing, Visualization, Funding
665 acquisition. **Yulin Deng:** Methodology, Software, Data Curation, Writing -
666 Original Draft. **Qiuxu Yi:** Methodology, Software, Data Curation. **Jun Yi:**
667 Conceptualization, Supervision, Project administration, Writing - Review &
668 Editing.

669 **Author Agreement**

670 Dear Editor: We declare that this manuscript entitled “SFD-ViT: Spatial-
671 Frequency Dual-Stream Transformer with Patch Embedding for Face Forgery
672 Detection” is original, has not been published before and is not currently be-
673 ing considered for publication elsewhere.

674 We confirm that the manuscript has been read and approved by all
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678 We understand that the Corresponding Author is the sole contact for
679 the Editorial process. He is responsible for communicating with the other
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681 **Declaration of competing interest**

682 The authors declare that they have no known competing financial inter-
683 ests or personal relationships that could have appeared to influence the work
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